EPSR-test-ex46 bonds: mean intramolecular distances

Program(s): bonds | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `bonds` (`bondsclus.f`), which checks the intramolecular geometry of an EPSR configuration [1]: for each run of consecutive molecules of the same type it writes the matrix of mean atom-atom distances (upper triangle) and their r.m.s. spreads (lower triangle) to a `.bonds` file. This is how a user verifies that the flexible molecules keep a sensible geometry during a refinement.

Code exercised

Fortran (original, fortran_src/):

- `bondsclus.f`: `bondsclus` (distance averages and spreads per run of identical molecules), main program (command line, file search)
- `ato_read_and_save.f`: `minitc` (reading the `.ato`)

Rust (port, rust/epsr-core/src/):

- `tools/bonds.rs`: `run` (averages, spreads, table output)
- `ato.rs`: `State::minitc`; `tools/groupc_common.rs`: `command_text`, `get_full_filename`

Physical system

The EPSR-test DMSO/water box (200 DMSO + 400 water molecules, 3200 atoms, cubic 33.1213 Å, 300 K): mean S-C 1.83 Å, S-O 1.55 Å, C-H 1.11 Å, water O-H 0.97 Å and H-H 1.55 Å, with spreads of 0.05-0.1 Å; distances from a methyl H to O or to the other methyl group spread by 0.3-0.7 Å, reflecting the methyl rotations. Also the single `DMSO.ato` and `WaterMolecule.ato` templates (spread 0), and `alternating.ato`: 7 molecules D D W W D W D in a 25 Å box.

Inputs

EPSR-test `.ato` files; `alternating.ato` made of randomly rotated, slightly distorted copies of DMSO and water by a helper script (a one-off script that was not kept; the generated file is the input). Commands `bonds 200DMSO400Water.ato`, `bonds DMSO` (no extension), `bonds WaterMolecule.ato`, `bonds alternating.ato`.

How the accuracy was verified

`commands.txt` (4 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Files compared: `200DMSO400Water.bonds` (two tables, averages over 200 and 400 molecules), `DMSO.bonds`, `WaterMolecule.bonds`, and `alternating.bonds` with five tables (a new table each time the molecule type changes: DD, WW, D, W, D). Compared against the original Fortran binary.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.